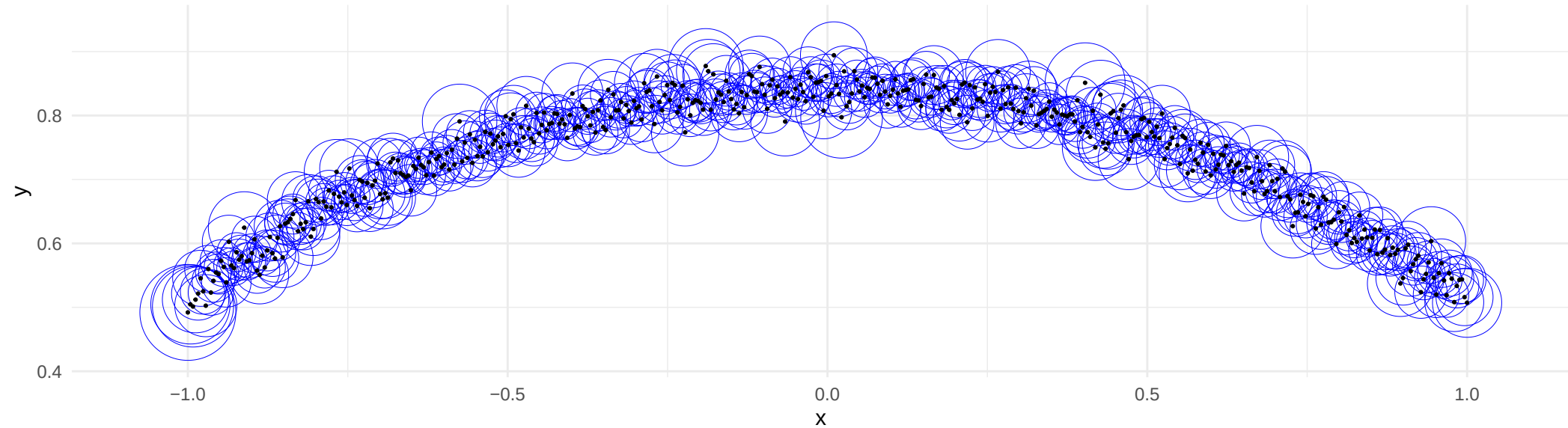
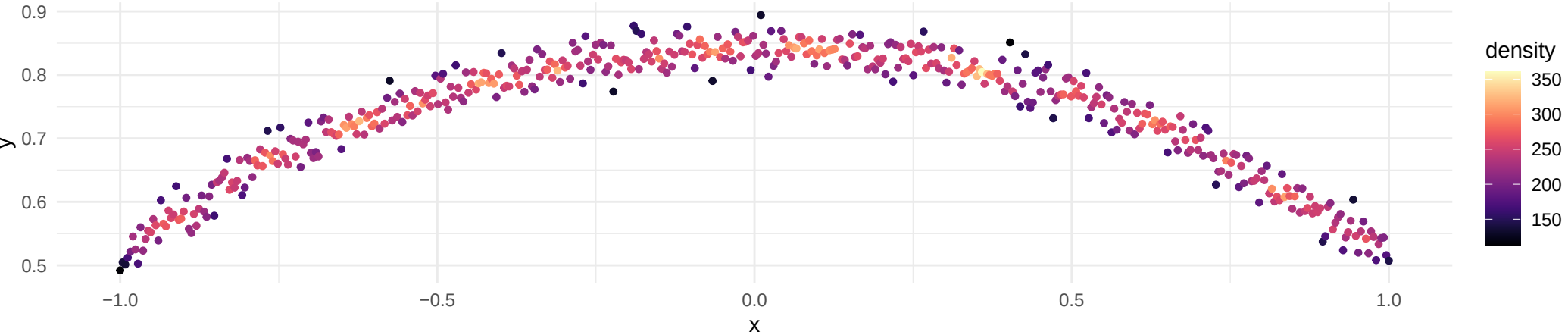


All Adaptive Spheres

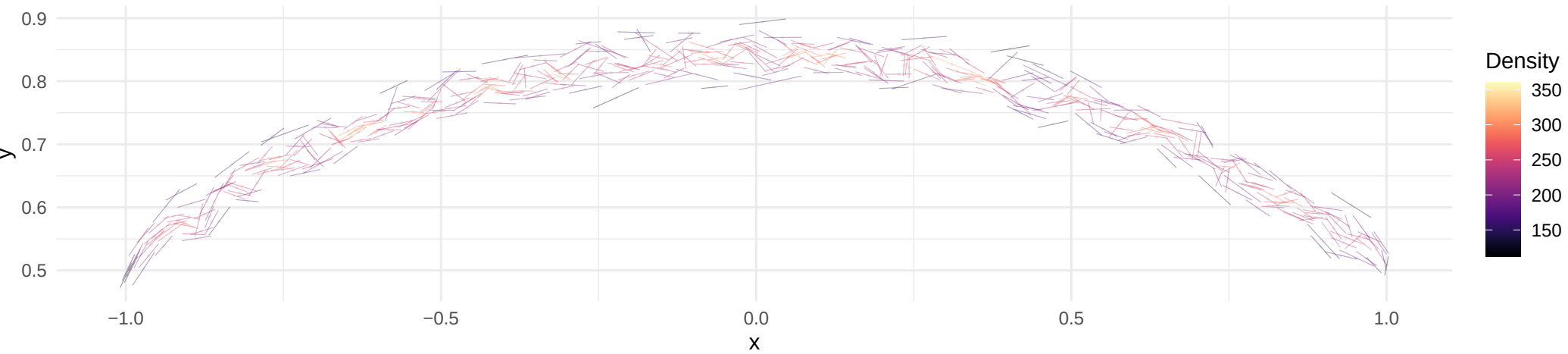
Visualizing local neighborhood size (radius) for every point.



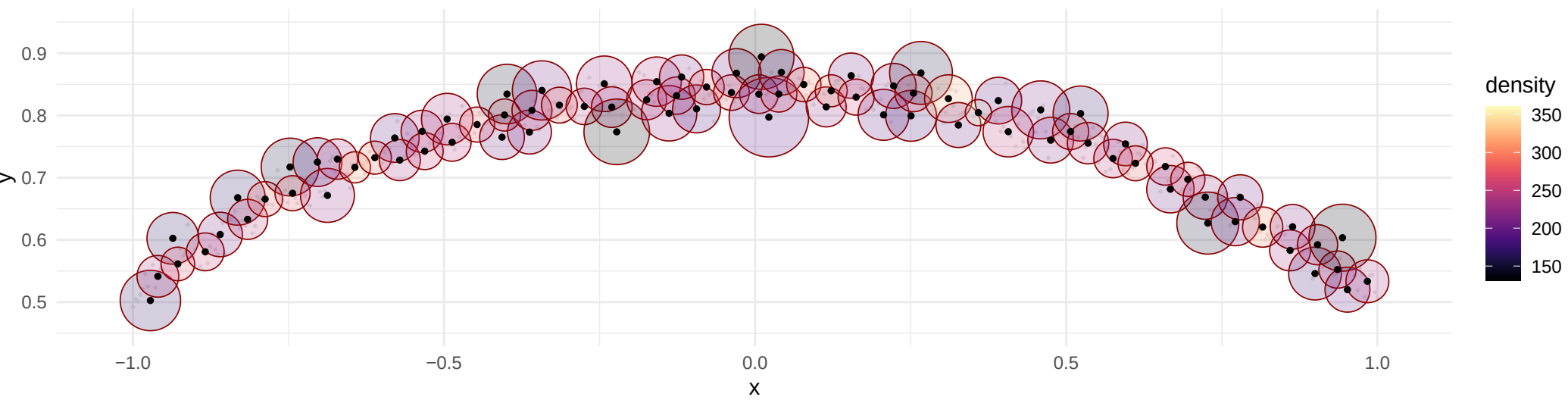
Point Density Distribution



Global Tangent Field (PC1)

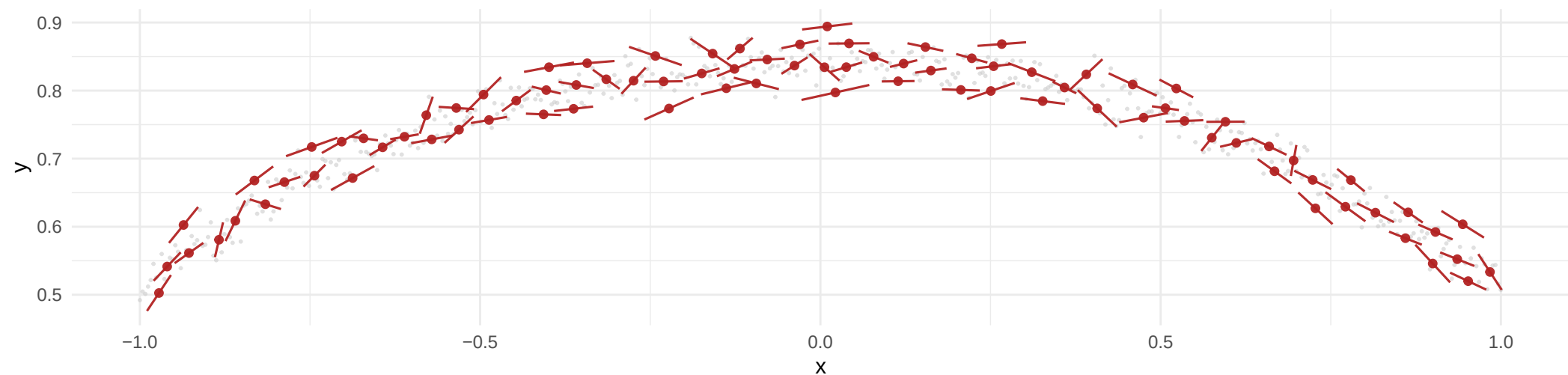


Atlas Selection (Anchors)



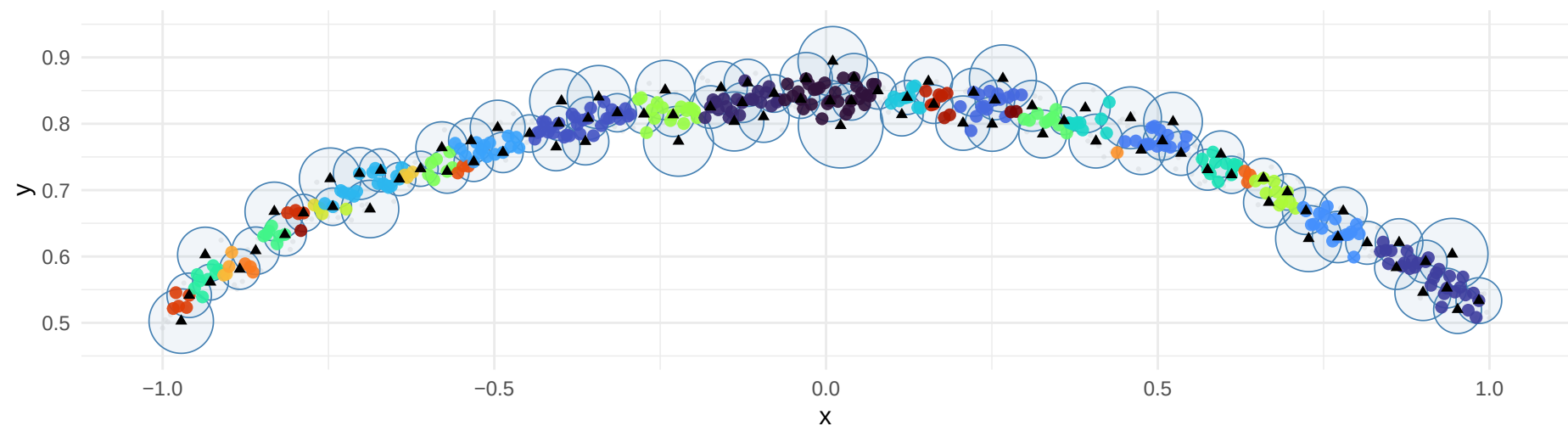
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



## Intersection Regions (2+ anchors)

Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



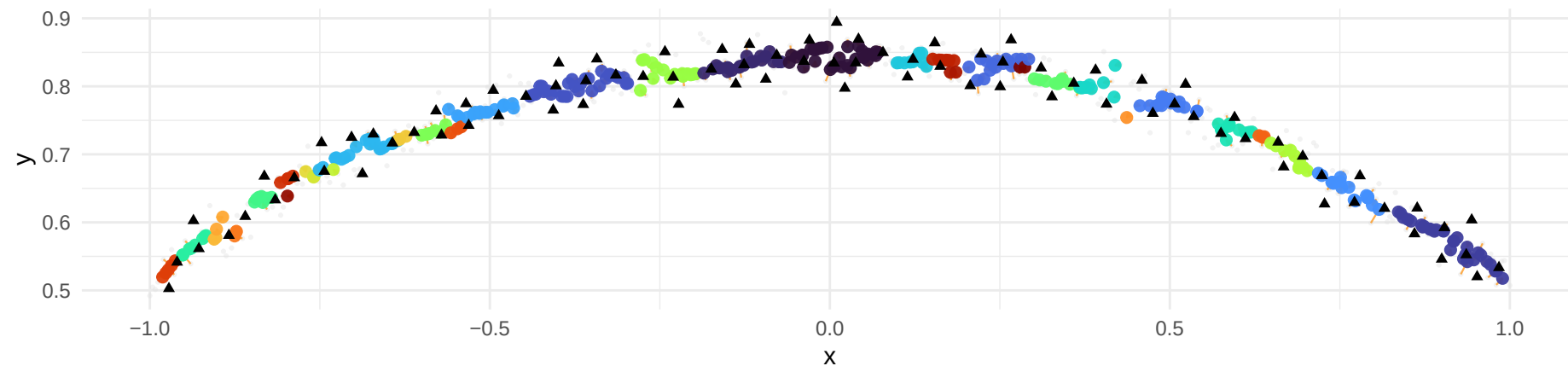
Parameters: k\_min = 10, g\_threshold = 3.0, o\_max = 0.3, o\_min = 0.10, stability = 0.90, scale\_factor = 2.5, max\_iter = 50, conv\_tol = 0.01

## Region

- |    |    |
|----|----|
| 1  | 18 |
| 2  | 19 |
| 3  | 20 |
| 4  | 21 |
| 5  | 22 |
| 6  | 23 |
| 7  | 24 |
| 8  | 25 |
| 9  | 26 |
| 10 | 27 |
| 11 | 28 |
| 12 | 29 |
| 13 | 30 |
| 14 | 31 |
| 15 | 32 |
| 16 | 33 |
| 17 | 34 |

# Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors

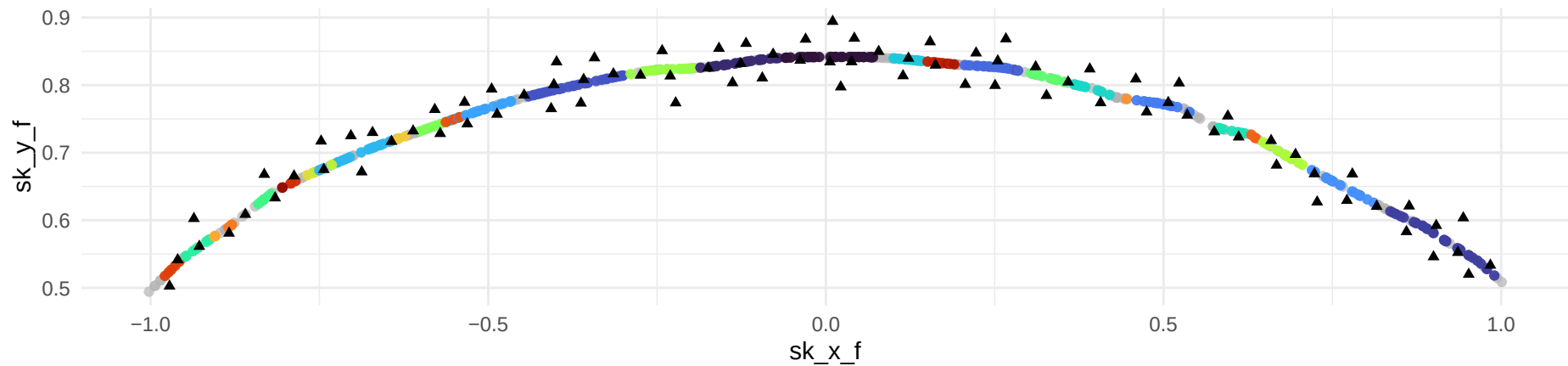


## Region

- |    |    |
|----|----|
| 1  | 18 |
| 2  | 19 |
| 3  | 20 |
| 4  | 21 |
| 5  | 22 |
| 6  | 23 |
| 7  | 24 |
| 8  | 25 |
| 9  | 26 |
| 10 | 27 |
| 11 | 28 |
| 12 | 29 |
| 13 | 30 |
| 14 | 31 |
| 15 | 32 |
| 16 | 33 |
| 17 | 34 |

# Final Stitched Backbone (Converged)

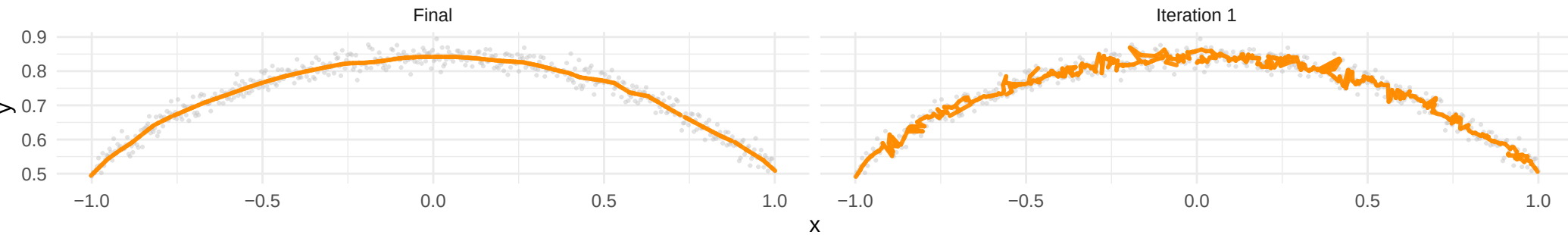
Final stitched points



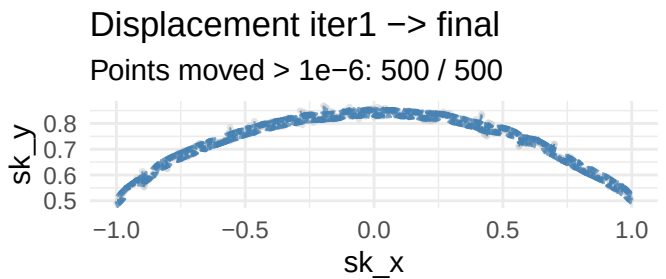
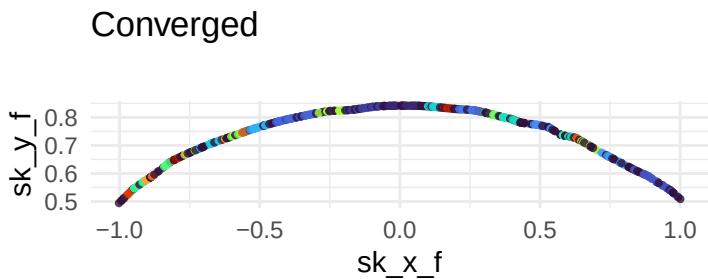
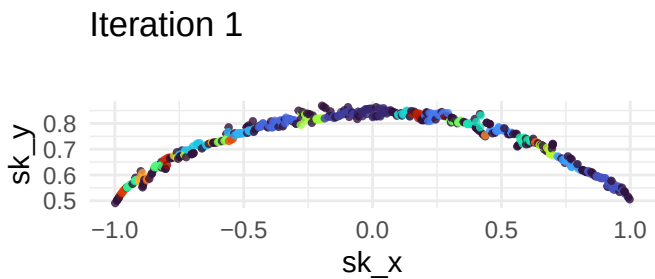
## Region

- |    |    |
|----|----|
| 1  | 18 |
| 2  | 19 |
| 3  | 20 |
| 4  | 21 |
| 5  | 22 |
| 6  | 23 |
| 7  | 24 |
| 8  | 25 |
| 9  | 26 |
| 10 | 27 |
| 11 | 28 |
| 12 | 29 |
| 13 | 30 |
| 14 | 31 |
| 15 | 32 |
| 16 | 33 |
| 17 | 34 |

Backbone Edges



Convergence: iteration 1 vs final



Parameters: k\_min = 10, g\_threshold = 3.0, o\_max = 0.3, o\_min = 0.10, stability = 0.90, scale\_factor = 2.5, max\_iter = 50, conv\_tol = 0.01